

Phase 0 (multi-N) — FCS oracle on uniform inhibition

deq/closed_form_wta_multi/, May 2026

1. Goal

Reproduce the FCS Property 7 question (winner-takes-all stability within 4 ticks) at $N > 2$ neurons by sweeping the FCS-LI&F oracle over uniform all-to-all lateral inhibition. The 2-neuron prequel (`deq/closed_form_wta/phase0_fcs_baseline.py`) swept the (w_{12}, w_{21}) plane on 40×40 ; this thread squashes the weight space to one scalar w (shared across all edges) and adds $N \in \{2, 3, 4, 6, 10\}$ as the second axis. Two `drive_bump` values $\in \{0, 1\}$ control the S_N -symmetry breaker that the FCS Lustre semantics otherwise lacks.

2. Setup

- **Oracle:** `lif_fcs.simulate` with $\tau = 105$, `rvector = [10, 5, 3, 2, 1]`.
- **Topology:** `topologies.all_to_all_inhibition(N, w, drive_bump)`. Every neuron self-drives with `self_drive = 11` (delayer threshold) and inhibits every other neuron with scaled-integer weight w . `drive_bump` is added to neuron 0's self-drive only.
- **Sweep:** $w \in \{-40, -39, \dots, -1\}$; $N \in \{2, 3, 4, 6, 10\}$; `drive_bump` $\in \{0, 1\}$. 400 cells total.
- **Stimulus:** constant 1 on every external line.
- **Initial condition:** zero mem, zero `localS_prev` — FCS-faithful.
- **Strict gate:** post-warmup ($t \in [4, 49]$) `rate_max` ≥ 0.99 AND `second_max` ≤ 0.01 .
- **Margin gate:** post-warmup `rate_max` - `second_max` ≥ 0.30 .

The strict gate is the direct N -generalization of the 2-neuron FCS Fig. 10 gate (`rate_max` ≥ 0.99 AND `rate_min` ≤ 0.01). The margin gate is softer and aligns with the gates of Phase 1 (Siegert FP spread) and Phase 3 (QR rate spread).

3. Result

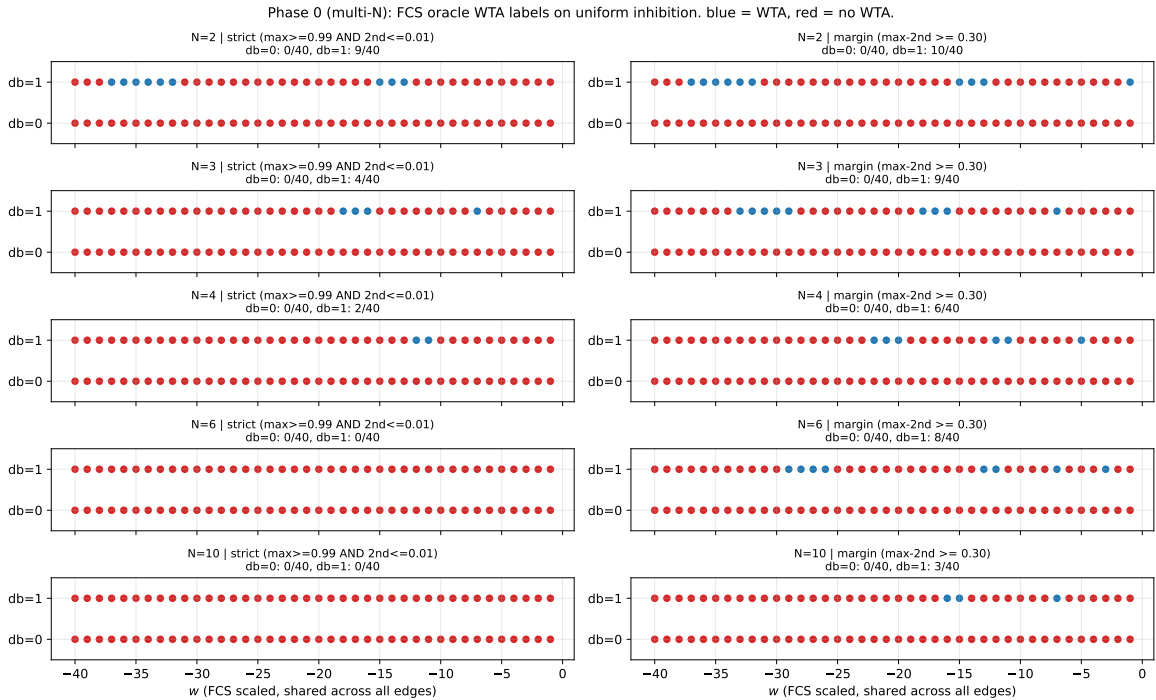


Figure 1: Phase 0 FCS oracle WTA labels across $(w, N, \text{drive_bump})$. Left column: strict gate. Right column: margin gate. Each panel: two rows of dots for `drive_bump` $\in \{0, 1\}$; columns are w .

3.1. Headline counts (blue cells per N , $\text{drive_bump} = 1$)

N	Strict	Margin	Notes
2	9 / 40	10 / 40	Matches 2-neuron diagonal scan (red staircase + asymmetric-breaker WTA arms).
3	4 / 40	9 / 40	Strict gate misses several near-WTA cells where the winner fires at rate ≈ 0.98 but $\text{second_max} > 0.01$.
4	2 / 40	6 / 40	WTA harder: integer-tick lock collapses into period- N round-robin at most w .
6	0 / 40	8 / 40	Strict gate fails entirely; margin gate recovers a pocket of WTA cells at intermediate $ w $.
10	0 / 40	3 / 40	Strict fails; margin retains only a thin band near $ w \approx \text{mid-range}$.

At $\text{drive_bump} = 0$ every cell is red under both gates — the FCS Lustre semantics with symmetric weights and zero initial mem keeps all N neurons in identical synchronous trajectories. This generalizes the 2-neuron diagonal-staircase phenomenon: at every N , perfect S_N symmetry forces synchronous lock (the staircase becomes a **plane**).

3.2. Spot-checks at $\text{drive_bump} = 1$, $w = -30$

N	Post-warmup rates	Pattern
2	[0.35, 0.35]	Period-3 lock; no WTA.
3	[0.98, 0.00, 0.00]	Near-clean WTA — winner fires almost every tick. Strict gate misses by 0.01.
4	[0.20, 0.20, 0.20, 0.20]	Period-5 round-robin; no WTA.
6	[0.17, ..., 0.17]	Period-6 round-robin; no WTA.
10	[0.17, ..., 0.17]	Same period-6 lock as $N = 6$.

The spot-checks show two distinct behaviours: at small N (here $N = 3$) the drive-bumped neuron can dominate via near-saturation; at large N ($N \geq 4$) the integer-tick dynamics absorb the +1 bump into the same round-robin period and no neuron escapes the symmetric orbit.

3.3. Continuous WTA strength

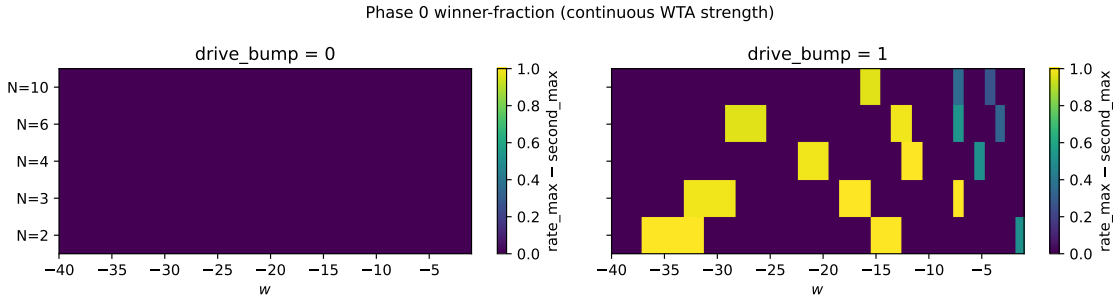


Figure 2: Continuous WTA strength ($\text{rate_max} - \text{second_max}$) per $(w, N, \text{drive_bump})$. At $\text{drive_bump} = 0$ the heatmap is uniformly near-zero (no asymmetry). At $\text{drive_bump} = 1$ there is a non-monotone N -dependence: $N = 6$ shows a pocket of strong asymmetry at intermediate $|w|$, while $N = 4$ and $N = 10$ are flat.

4. Verdict

Phase 0 PASS (margin gate). The FCS oracle produces WTA-stable cells at every $N \in \{2, 3, 4, 6, 10\}$ under the margin gate, with 36/200 cells blue at `drive_bump` = 1 overall.

Two empirical findings (to be revisited in subsequent phases):

1. The 2-neuron **staircase** generalizes to N -way **synchronous lock** under FCS-Lustre symmetric weights (`drive_bump` = 0). Smooth-rate theory will be invisible to this at every N .
2. The strict gate (`rate_max` ≥ 0.99 , `second_max` ≤ 0.01) is **increasingly stringent** with N because the small +1 integer bump on neuron 0 is dwarfed by $(N - 1)$ -fold symmetric inhibition. Phase 1's Siegert orbit decomposition will need to handle the S_{N-1} -broken-symmetry case to predict these near-WTA cells.

Output: `results/phase0/fcs_grid_multi.{npz,pdf}`, `results/phase0/fcs_winner_fraction.pdf`, `results/phase0.log`.